

How to Get the GCP_SA_KEY for GitHub Actions

This guide explains how to create a Google Cloud service account key and add it as a GitHub Actions secret to enable automated deployments to Cloud Run.

Step 1: Create a Service Account

Run the following command to create a dedicated service account for GitHub deployments:

```
gcloud iam service-accounts create github-deployer \
  --project=stock-auto-trader-495209 \
  --display-name="GitHub Deploy Bot"
```

Step 2: Grant It the Required Roles

The service account needs four IAM roles. Run each command:

```
SA="github-deployer@stock-auto-trader-495209.iam.gserviceaccount.com"

gcloud projects add-iam-policy-binding stock-auto-trader-495209 \
  --member="serviceAccount:$SA" --role="roles/run.admin"

gcloud projects add-iam-policy-binding stock-auto-trader-495209 \
  --member="serviceAccount:$SA" --role="roles/storage.admin"

gcloud projects add-iam-policy-binding stock-auto-trader-495209 \
  --member="serviceAccount:$SA" --role="roles/cloudbuild.builds.editor"

gcloud projects add-iam-policy-binding stock-auto-trader-495209 \
  --member="serviceAccount:$SA" --role="roles/iam.serviceAccountUser"
```

Why `serviceAccountUser` ? Cloud Run acts as a service account when deploying, so the deployer must have permission to impersonate it.

Step 3: Download the JSON Key

```
gcloud iam service-accounts keys create ~/github-deployer-key.json \
  --iam-account="github-deployer@stock-auto-trader-495209.iam.gserviceaccount.com"
```

This creates a file that looks like:

```
{
  "type": "service_account",
  "project_id": "stock-auto-trader-495209",
  "private_key_id": "abc123...",
```

```
"private_key": "-----BEGIN RSA PRIVATE KEY-----\n...",
"client_email": "github-deployer@stock-auto-trader-
495209.iam.gserviceaccount.com",
...
}
```

Step 4: Add the Key as a GitHub Secret

1. Go to your GitHub repository
2. Navigate to **Settings** → **Secrets and variables** → **Actions**
3. Click **New repository secret**
4. Set the name to `GCP_SA_KEY`
5. Paste the **entire raw JSON content** of `~/github-deployer-key.json` as the value

Note: The `google-github-actions/auth@v2` action accepts raw JSON directly — no base64 encoding needed.

Also add a second secret:

Name	Value
GCP_PROJECT_ID	stock-auto-trader-495209

Step 5: Delete the Local Key File

```
rm ~/github-deployer-key.json
```

Never commit this file to git. The secret is now stored securely in GitHub and the local copy is no longer needed.

Summary

Step	Action
1	Create service account <code>github-deployer</code>
2	Grant 4 IAM roles (<code>run.admin</code> , <code>storage.admin</code> , <code>cloudbuild.builds.editor</code> , <code>iam.serviceAccountUser</code>)
3	Download JSON key to local disk
4	Paste JSON content into GitHub secret <code>GCP_SA_KEY</code>
5	Delete local key file

Once done, trigger the deploy workflow manually from **GitHub** → **Actions** → **Deploy to Cloud Run** → **Run workflow**.